

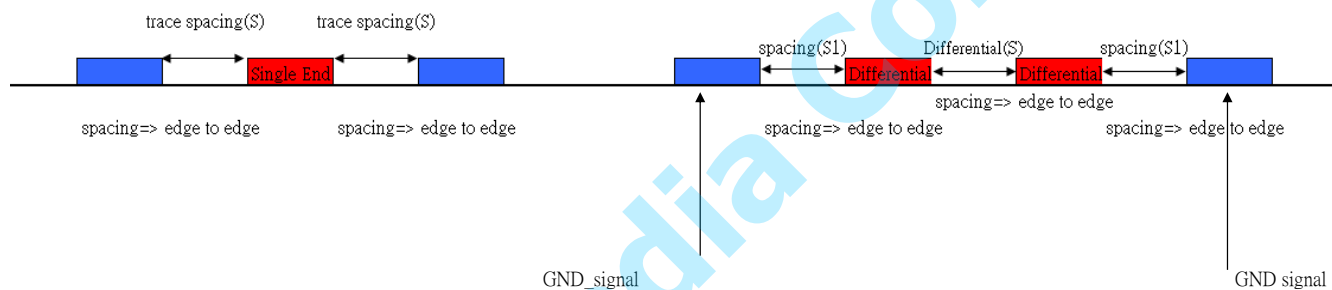
ASM2464PD PCB Guideline

ECN

Date	Revision	Item	Description	Explanation
2022/5/13	1.0		Release	

PCB Stack-up

			Single end +/-15%		Differential +/-10%	
Thickness	1.58mm+/-10%		Trace Width / impedance		Trace Width spacing / impedance	
Layer	lyr type	Thickness	50ohm	90 ohm	95 ohm	85 ohm
S/M	S/M	0.7				
L1-COMP	0.5oz+plating	1.4	4 / 50.44	4 5 / 91.67	4 7 / 95.94	4 4 / 85.46
	PP 1080MR	2.62				
L2-GND	1oz	1.3				
	CORE	48.4				
L3-VCC	1oz	1.3				
	PP 1080MR	2.62				
L4-SOLD	0.5oz+plating	1.4	4 / 50.44	4 5 / 91.67	4 7 / 95.94	4 4 / 85.46
S/M	S/M	0.5				
Thickness(mil)		60.24				
mm		1.530096				



USB4 20Gbps

UTXPIN/IO/L_URXPN/IO/L Differential Pair Guidelines

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Layer	Microstrip	
Characteristic Trace Impedance	Difference Pair Impedance = 85 ohm +/- 10 ohm	Shielding is needed for USB 4 differential pairs for signal integrity and EMC performance as below. A radial bend is preferred for 20Gbps.
Differential (S1:W:S:W:S1)	30 : 4 : 4 : 4 : 30 (S1 > 8 h)	Each distance 40mil with ground via hole for guard trace as below.
Serpentine Spacing		No Serpentine
VIA hole Max.	<= 2	
Total Length	< 1.5" for type C SMT connector, ASAP (prefer 0.5"~1.2") or Maximum insertion loss < 4dB@10GHz.	maintain loss, crosstalk and fiber weave effect. Shielding is needed for USB 4 differential pairs for signal integrity and EMC performance. TX0TX1 length match < 1000 mil, RX0RX1 length match < 1000mil for lane to lane skew.
Series Capacity	C_TX<0.22uF (Size 0201 / X7R) ; C_RX<0.33uF (Size 0201 / X7R) R_DN= 220K ohm (Size 0201)	The Cap close to Connector for UTXPN; RX AC coupling Cap and resistor pull down for short protection. (Component size should be 0201 on USB4.)
Differential Pair PIN Length Matching	< 5 mil	Every segment intra pair skew: < 1 mil

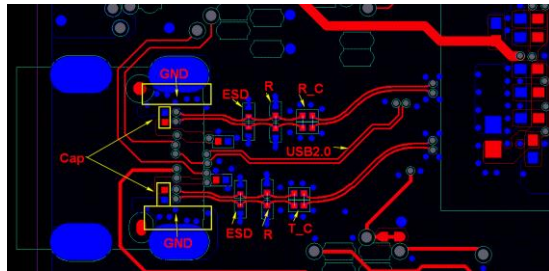
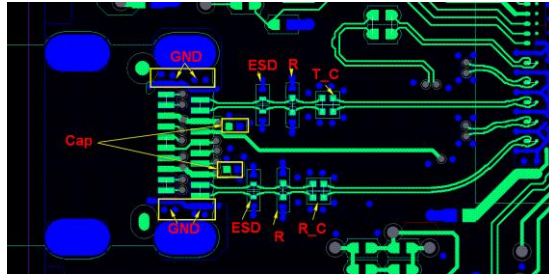
USB2.0 UDPIN Differential Pair Guidelines

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Characteristic Trace Impedance	Difference Pair Impedance = 85 ohm +/- 10%	
Differential (S1:W:S:W:S1)	15 : 4 : 4 : 4 : 15	
Serpentine Spacing	15 mil	
VIA hole Max.	< 4	
Total Length	< 4"	
Differential Pair Length Matching	< 20 mil	

Parameter	Guidelines	Guidelines	Guidelines
Signal Name	UREXT	VBUS	VCC3.3/VDD/VCC
Topology	Point to Point	Point to Point	Point to Point
Signal End (W:S)	10 : 7	6 : 8	15 : 8
Total Length	ASAP	10"	ASAP

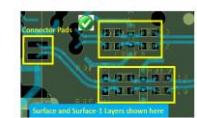
Refer to USB4 system design guidelines

<https://www.asmedia.com.tw/Upload/Doc/USB4%20System%20Design%20Guidelines.pdf>
TYPE C only SMT type)



Layout Design Considerations/Examples - 5/7

- Void under SMT pads for better SI. The larger the pad size, the more important this is



- **Single-ended voids** are recommended for most cases/components

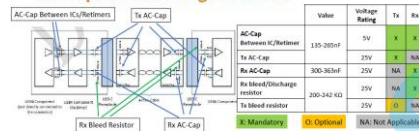


- **Differential voids** for larger pads and/or thin dielectric height ($\leq 60\mu\text{m}$)
- These voids can over-compensate (increase impedance too much). So analysis/3D modeling may be needed

- Ensure that layer under the void (surface-2) is not a source of noise, e.g. power plane. It's best to have ground on surface-2 under the void

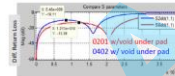
USB4 System Design 4-2022

AC-Caps and Discharge Resistors



	Value	Voltage Rating	Tx	Rx
AC-Cap Between ICs/Repeater	150-200pF	5V	X	X
Tx AC-Cap	25V	X	NA	
Rx AC-Cap	300-350pF	25V	NA	X
Rx bleed/Discharge resistor	25V	NA	X	
Tx bleed resistor	200-242 kΩ	25V	O	NA
			X: Mandatory	O: Optional
			NA: Not Applicable	

- Smaller size components (e.g. 0201) will have smaller parasitic and therefore better return loss than larger ones (0402). Consider using 0201 components



PCIe Gen4 16Gbps

PTXPx/Nx, PRXPx/Nx, Differential Pair Routing Guidelines

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Layer	Microstrip	
Characteristic Trace Impedance	TX&RX Impedance Difference Pair = 85 ohm +/- 10%	
Differential (S1:W:S:W:S1)	30 : 4 : 4 : 4 : 30(S1 > 8 h)	
Serpentine Spacing		
Series Capacity	Cs=0.22uF (Size 0402 / X7R or Size 0201(max. length))	The AC-couple capacitor close to connector for PETXP/N
VIA hole Max.	<= 2 for connector ; <= 4 for down device or chip	
Total Length	< 3" for connector, ASAP. 1.5" < Down device(on board device or chip to chip) < 4.5"	
Differential Pair P/N Length Matching	< 5 mil	

XO/XI

Parameter	Guidelines
Signal Name	XO /XI
Characteristic Trace Impedance	Impedance Difference Pair = 85 ohm +/- 15%
Differential (S1:W:S:W:S1)	30 : 4 : 4 : 4 : 30
Total Length	< 0.5" ,ASAP

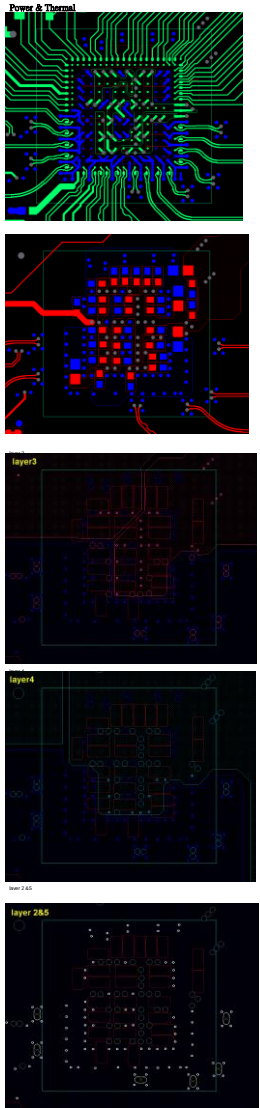
Other

Parameter	Guidelines
Signal Name	VDD/VCCLP/VCCLU/VCCM/VCCX
Topology	Point to Point
Single End (W:S)	15 : 8
Total Length	ASAP

PECLK

PE CLKP/N

Parameter	Guidelines	Notes
Signal Reference	Ground	
Signal Name	PECLKP/N	
Layer	Microstrip	
Characteristic Trace Impedance	Impedance Difference Pair = 85 ohm +/- 15%	
Differential (S1:W:S:W:S1)	30 : 4 : 4 : 4 : 30(S1 > 8 h)	
Total Length	< 6" for connector ,< 8" for chip pad, ASAP.	
Differential Pair P/N Length Matching	< 5 mil	



ASMedia Confidential

SPIDI/SPICK/SPIDO/SPICS/I2C_DATA/I2C_CLK

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Characteristic Trace Impedance (Single-Ended)	50 ohm +/- 15%	
Trace Width	5 mils	
Trace Spacing	10 mils	
Total Length	< 10"	

CC1/CC2

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Trace Width	8 mils	
Trace Spacing	15 mils	
Total Length	< 10"	

GPIO/Strapping/UART_TX/UART_RX

Parameter	Guidelines	Notes
Topology	Point to Point	
Signal Reference	Ground	
Characteristic Trace Impedance (Single-Ended)	50 ohm +/- 15%	
Trace Width	5 mils	
Trace Spacing	8 mils	
Total Length	< 10"	